FINAL PROJECT · ADVANCED TECHNOLOGIES

Smart Parking Lot

Intelligent vehicle access control system with clean architecture, SOLID principles and IoT hardware support

- C# 14 / .NET 10
- Clean Architecture
- Arduino
- SQLite

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CONTEXT

Problem Statement

Traditional parking lots rely on **manual processes and human supervision** to record vehicle entry and exit, determine space availability, and control access.

This creates **operational bottlenecks**, registration errors, long wait times and a poor user experience, along with unnecessary staffing costs.

The absence of an automated system makes **real-time monitoring**, vehicle traceability and data-driven decision making impossible.

Inefficient Manual Registration

Without automation, vehicle entry and exit depends on operators and error-prone physical forms.

No Real-Time Visibility

Drivers don't know how many spaces are available before entering, causing congestion and wasted time.

Lack of Traceability

No plate-based access history exists, making audits, accurate billing and usage reporting impossible.

REQUIREMENTS

Use Cases Table

Use Case	Sensor(s) / Peripheral(s)	Trigger Condition	Expected System Reaction	Data to Persist	Done?
Parking Occupancy Detection	IR Sensor, ArduinoSerialBridge, LED	Arduino sends a SpotSensorReading via IArduinoReader	Captures reading, updates spot state & parking availability	Spot ID, occupancy status, timestamp	✓ Yes
Entry Control (LPR)	Camera Module, ArduinoSerialBridge, Servo Motor	Arduino sends GateSensorReading with detected plate	Creates EntryRequest, checks capacity, opens gate if approved	Plate, entry timestamp, approval status	✗ No
Exit Control (LPR)	Camera Module, ArduinoSerialBridge, Servo Motor	Arduino sends GateSensorReading at exit gate	Creates ExitRequest, releases spot, opens gate	Plate, exit timestamp, released spot ID	✗ No
Full Capacity Alert	IR Sensors, ArduinoSerialBridge, LED	CapacityService finds no spots after processing SpotSensorReading	Generates alert via AlertService, blocks new EntryRequests	Alert ID, type, message, timestamp	✗ No

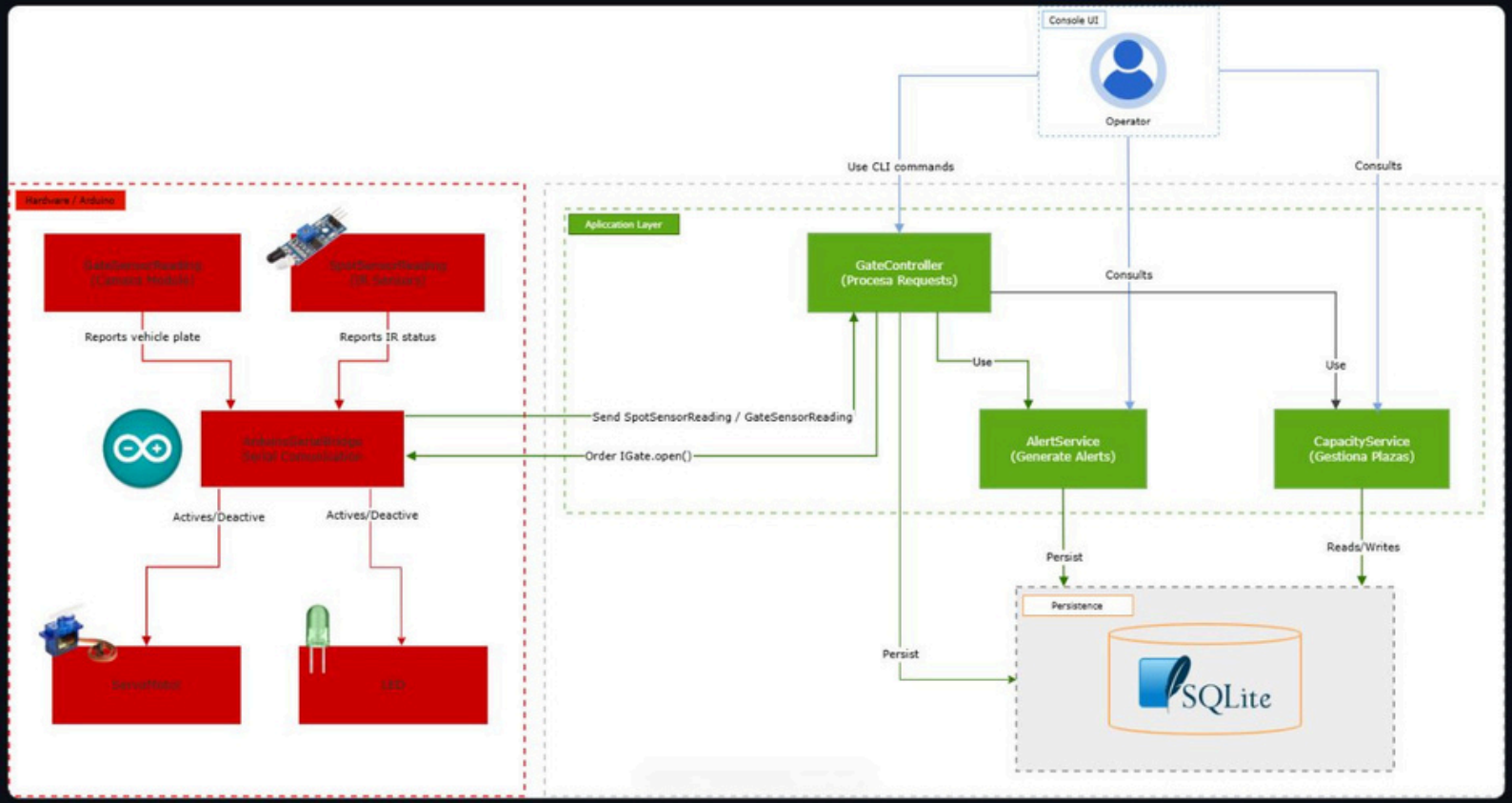
HARDWARE

Materials Matrix

Component	Qty	Role in the Project
Arduino Uno	2	Processes sensor logic and dispatches commands to actuators
IR Sensor (Infrared)	8	Detects vehicle presence in each parking spot
Servo SG90	2	Opens and closes the physical access barriers (entry & exit gates)
Camera OV7670	2	Captures vehicle plates for License Plate Recognition (LPR) at entry & exit
LEDs (Red / Green)	2	Visual guidance for driver — indicates spot availability
Jumper Wires	30+	Hardware interconnection between all components

DESIGN

System Architecture

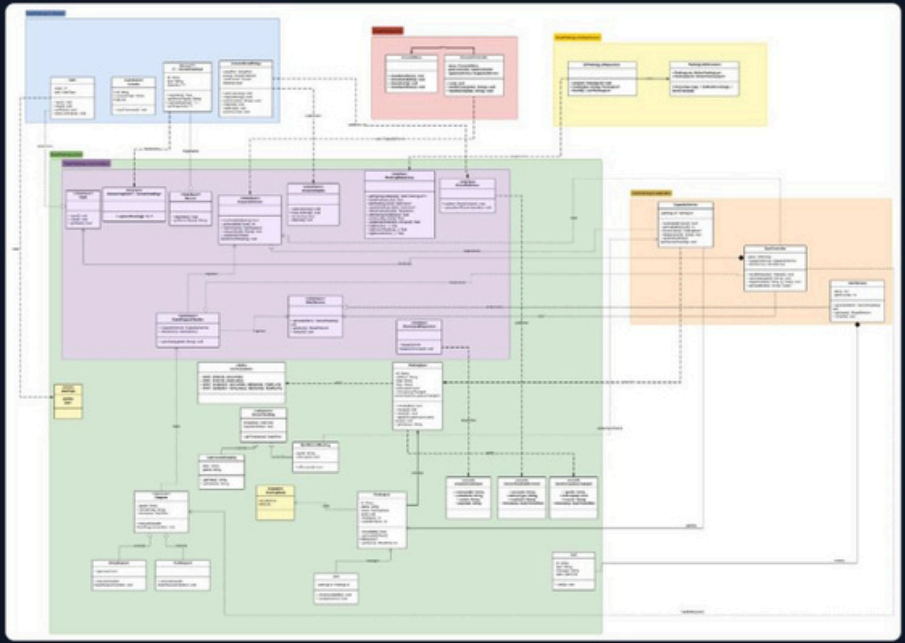


MODELING

UML Class Diagram

Represents the static structure of the system — classes, interfaces and their relationships across the four-layer clean architecture.

- **Core**
  Entities & interfaces — zero dependencies
- **Application**
  Controllers and application services
- **Hardware**
  Sensors, actuators and physical gate
- **CLI**
  Composition root & dependency injection



SOFTWARE DESIGN

SOLID Principles Applied

S

Single Responsibility

Each class has one clearly defined responsibility within the system.

```
ParkingSpot → state
GateController → flow
CapacityService → capacity
```

O

Open / Closed

Abstractions allow extending the system without modifying existing code.

```
IGateRequestHandler
allows new implementations
```

L

Liskov Substitution

Concrete implementations are interchangeable with their interfaces without breaking the system.

```
Gate : IGate
Sensor : ISensor
are substitutable
```

I

Interface Segregation

Small, focused interfaces — no class is forced to implement methods it doesn't need.

```
ICapacityService
IAlertService
IGate · ISensor
```

D

Dependency Inversion

High-level modules depend on abstractions defined in Core, not on concrete implementations.

```
Core → 0 deps
Application → Core
Hardware → Core
```

INTERFACE

Application Menu

The application exposes an interactive console menu (CLI) to operate the parking lot, monitor sensors and query history from the terminal.

Options marked with # are features under development not yet implemented in the current demo.

SmartParkingLot.Cli

```
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‖ Smart Parking Lot — Menú ‖
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Parking Lot : Main Parking Lot (LOT-001)
Mode : Standard
Spots : 3 total | 2 available
# → Not implemented in this demo

1. # Request vehicle entry
2. # Request vehicle exit
3. # Update spot status (manual sensor)
4. View parking lot status
5. # View vehicle history
6. View sensor readings
7. # View device actions
8. Live monitoring (Arduino)
9. View spot status
0. Exit
```

ROADMAP

Future Implementations

1

Vehicle Entry Request

Integration with License Plate Recognition (LPR) to automatically authorize entry, log the event and assign an available spot in real time.

2

Vehicle Exit Request

Automatic exit detection via gate sensor, spot release, stay duration calculation and payment receipt generation.

5

Vehicle History

Full access history by plate: dates, entry/exit times, assigned spots and approval status for each request.

7

Device Actions Log

Recording and display of the action log for each physical actuator (gates, LEDs), useful for hardware audits and fault diagnosis.

+

Desktop GUI

A visual Desktop layer added to the architecture, preserving the Dependency Inversion Principle by depending only on Core abstractions.

STACK

Technologies Used

C#

C# 14

Lenguaje principal

Modern syntax with Records, Pattern Matching, top-level statements and nullable reference types for compile-time safety.

Records · Pattern Matching · Nullable

.NET

.NET 10

Runtime & SDK

Cross-platform runtime. Solution with four independent projects interconnected via project references.

Multi-proyecto · DI Manual

Arduino

Hardware IoT

Serial communication with microcontroller for IR sensor readings and LED actuator control from C# via COM port.

Serial Bridge · Sensores IR · LEDs

SQLite

Persistencia

Serverless embedded database to persist parking lot state, request history, sensor readings and device actions.

Embebida · Sin servidor

Thanks!

Smart Parking Lot — Advanced Technologies

- C# 14 / .NET 10
- Clean Architecture
- SOLID & GRASP
- Arduino + SQLite